

Math 141, F11
Exam 3

Section:

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 4 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	28	
2	22	
3	30	
4	20	
Total	100	

Good Luck !

Problem 1. (28 points) Find each of the following requested limits using l'Hopital's Rule when applicable.

(a) $\lim_{x \rightarrow 1} \frac{x-1}{x^3-1}$

$$= \lim_{x \rightarrow 1} \frac{1}{3x^2} = \frac{1}{3}$$

(c) $\lim_{x \rightarrow \infty} x e^{-x}$

$$= \lim_{x \rightarrow \infty} \frac{x}{e^x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

(b) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$

$$= \lim_{x \rightarrow 0} \frac{2 \sin 2x}{2x} = \lim_{x \rightarrow 0} \frac{2 \cos 2x}{1} = 2$$

(d) $\lim_{x \rightarrow 0^+} x^{x^2}$ (Hint: use natural logarithm)

$$y = x^{x^2}$$

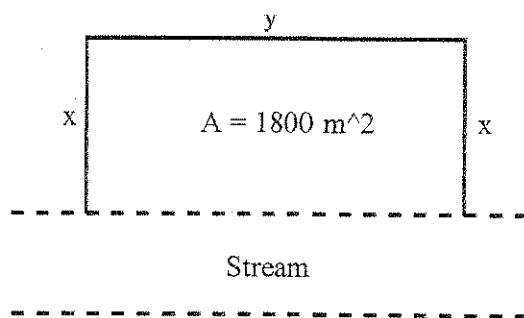
$$\ln y = x^2 \ln x$$

$$\lim_{x \rightarrow 0^+} x^2 \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{2}{x^3}}$$

$$= \lim_{x \rightarrow 0^+} -\frac{x^2}{2} = 0$$

$$\text{Thus } \lim_{x \rightarrow 0^+} x^{x^2} = e^{\lim_{x \rightarrow 0^+} \ln y} = e^0 = 1$$

Problem 2. (22 points) A rectangle field is to be bounded by a fence on three sides and by a straight stream on the fourth side, and the area of the field is required to be $A = 1800 \text{ m}^2$, see Figure below. Find the dimensions of the field that uses the shortest length of fencing.



Solution: $L = 2x + y$. $\rightarrow$ length of fencing

$$A = xy = 1800, \quad x \geq 0, y \geq 0$$

$$\Rightarrow y = \frac{1800}{x}$$

Thus

$$L = 2x + \frac{1800}{x}, \quad x \in [0, +\infty).$$

$$\frac{dL}{dx} = 2 - \frac{1800}{x^2} = 0$$

$$x^2 = \frac{1800}{2}$$

$$x^2 = 900$$

$$x = 30 \text{ m (Since } x \geq 0).$$

$$\text{Then } y = \frac{1800}{30} = 60 \text{ m}$$

Problem 3. (30 points) Evaluate each of the following indefinite/definite integrals.

$$\begin{aligned}
 \text{(a)} \quad & \int (x^7 - 2x^{-3} - \sqrt[5]{x}) dx \\
 &= \frac{x^8}{8} - 2 \cdot \frac{x^{-2}}{-2} - \frac{x^{5/4}}{5/4} + C \\
 &= \frac{x^8}{8} + \frac{1}{x^2} - \frac{4}{5} x^{5/4} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad & \int e^{-2x} dx \\
 & \xrightarrow{u=-2x} \int -\frac{1}{2} e^u du \\
 &= -\frac{1}{2} e^u + C \\
 &= -\frac{1}{2} e^{-2x} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad & \int 2x(x^2 + 3)^{10} dx \\
 & \xrightarrow{u=x^2+3} \int u^{10} dx \\
 &= \frac{1}{11} u^{11} + C \\
 &= \frac{1}{11} (x^2 + 3)^{11} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad & \int \frac{\cos t}{1 + \sin^2 t} dt \\
 & \xrightarrow{u=\sin t} \int \frac{1}{1+u^2} du \\
 &= \arctan u + C \\
 &= \arctan(\sin t) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(e)} \quad & \int_e^{e^2} \frac{1}{x \ln x} dx \\
 & u = \ln x \Rightarrow du = \frac{1}{x} dx \\
 & x = e \rightarrow u = \ln e = 1 \\
 & x = e^2 \rightarrow u = \ln e^2 = 2
 \end{aligned}$$

$$\begin{aligned}
 & \text{thus} \\
 & \int_e^{e^2} \frac{1}{x \ln x} dx \xrightarrow{u=\ln x} \int_1^2 \frac{1}{u} du \\
 &= [\ln|u|]_{u=1}^2 \\
 &= (\ln 2) - (\ln 1) \\
 &= \ln 2
 \end{aligned}$$

Problem 4. (20 points)

Let

$$v(t) = 2 \sin(\pi t)$$

(measured in feet per second) be the velocity of a particle moving along a coordinate line.

(a) Find the displacement of the particle during the time period $0 \leq t \leq 2$.

$$\begin{aligned} S &= \int_0^2 v(t) dt \\ &= \int_0^2 2 \sin \pi t dt \\ &= \left[-\frac{2}{\pi} \cos \pi t \right] \Big|_{t=0}^2 \\ &= \left(-\frac{2}{\pi} \right) - \left(-\frac{2}{\pi} \right) \\ &= 0 \end{aligned}$$

(b) Find the distance traveled by the particle during the same time period as (a).

$$\begin{aligned} d &= \int_0^2 |v(t)| dt \\ &= \int_0^2 |2 \sin \pi t| dt \\ &= \int_0^1 2 \sin \pi t dt + \int_1^2 (-2 \sin \pi t) dt \\ &= \left[-\frac{2}{\pi} \cos \pi t \right] \Big|_{t=0}^1 + \left[\frac{2}{\pi} \cos \pi t \right] \Big|_{t=1}^2 \\ &= (0) - \left(-\frac{2}{\pi} \right) + \left(\frac{2}{\pi} \right) - (0) \\ &= \frac{4}{\pi} \end{aligned}$$

Extra Credits. (5 points) Evaluate $\int \frac{x}{\sqrt{1-x^4}} dx$.

Solution: $u = x^2$

$$du = 2x dx \Rightarrow \frac{du}{2} = x dx$$

$$\int \frac{x}{\sqrt{1-x^4}} dx \stackrel{u=x^2}{=} \int \frac{1}{\sqrt{1-u^2}} \frac{du}{2}$$

$$= \frac{1}{2} \arcsin u + C$$

$$= \frac{1}{2} \arcsin(x^2) + C$$